

Applied Physics

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Assignment 1

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BSCS-1C

Introduction

Applied physics is an important branch of science that deals with the practical use of physical principles in different fields of technology and engineering. It helps us understand how the laws of physics can be used to develop useful devices, improve existing systems, and solve real-life problems. Many modern technologies, including computers, communication systems, industrial machines, and medical equipment, work on the basic principles of physics.

This report discusses the role of applied physics in five major areas: quantum mechanics, microelectronics, fiber optics, laser technologies, and medical technologies. It explains the basic physical principles behind these fields and describes how they are used in modern technology.

Quantum Mechanics

It is the branch of Physics that studies the behavior of Matter and energy at the atomic & subatomic levels. It explains several phenomena that cannot be fully understood by classical physics

Applications

- **Quantum Computing**

Quantum computers use Quantum bits or qubits which can be prepared in super position states. These properties may help solve certain problems more efficiently than conventional computers

- **Quantum key distribution**

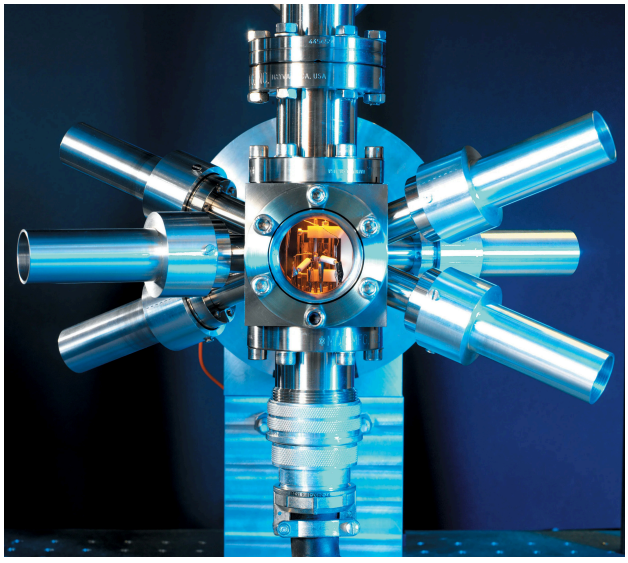
QKD uses quantum states, such as the polarization of photons to establish secure cryptographic keys

- **Atomic clocks**

Atomic clocks ~~use~~ measure the precise frequencies of atomic ~~transition~~ transitions. For example: cesium-133 atomic clocks use a specific atomic trasition to maintain highly accurate time, which is Important for GPS & other ~~sat~~ synchronization systems



Quantum Computer



Atomic Clock

Microelectronics

It deal with the design and manufacturing of very small components and circuits. It's an important part of Modern computers, smartphones, communication equipment and many other electronic devices.

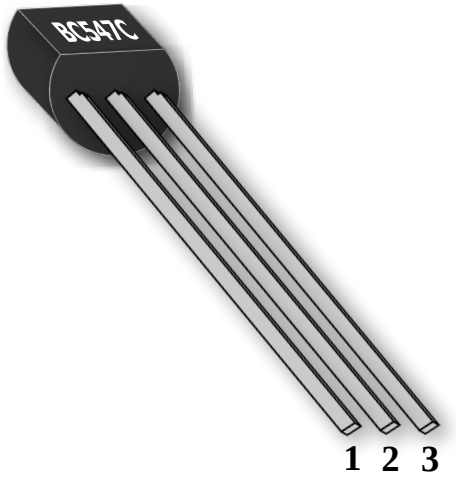
Applications

- Transistors

They act as electronic switches and can also amplify signals. MOSFETs are widely used in ICs while FinFETs use a three-dimensional channel structure to improve control over current and reduce leakage in small devices

ICs & Microprocessor

ICs contain a large number of electronic component on a small semiconductor chip. Modern Microprocessor use billion of transistors to perform calculation, process information and control different functions of electronic devices



Transistors



Apple's M5 Microprocessor

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Fiber Optics

It is the technology that uses thin strand of glass or plastic to transmit information in the form of light. It is widely used in internet connection, telecommunications, medical instrument and Industrial monitoring systems

Applications

- Telecommunication

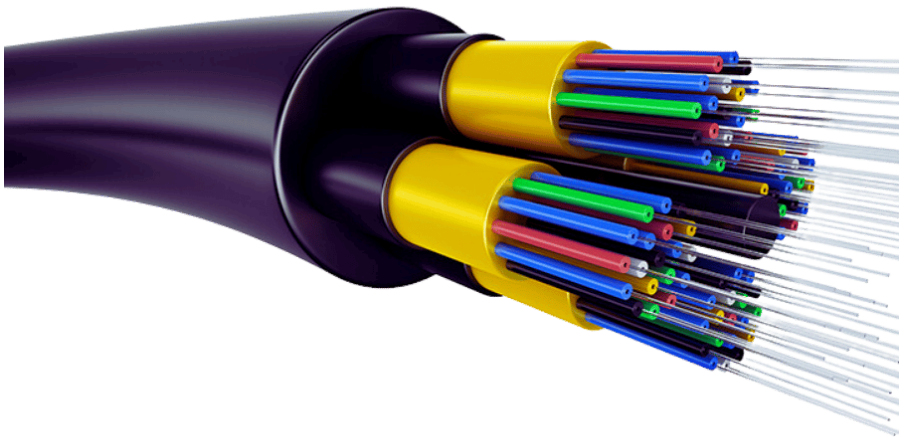
Optical fibers are used to transmit large amount of data over long distances. They form an Important part of internet Infrastructure, including those Under-sea cables

- Erbium-Doped Fiber Amplifiers

They amplify optical signals directly inside a fiber. They use Erbium doped material and energy supplied by a pump laser to strengthen signals without converting them into electrical signals

- Endoscopes

In medicine, flexible optical fibers are also used in endoscope to help doctors examine internal parts of body without requiring major surgical procedure



Industrial Fiber Optic Cable



Erbium-Doped Fiber Amplifier

Laser Technology

It is an device that produces a concentrated beam of light through a process known as simulated emission. Laser light is an useful because it can be highly directional, relatively narrow in wavelength and coherent

Application

- Industrial processing

  High power CO_2 and Fiber lasers are used for cutting, welding, engraving and Marking Materials

- Optical Storage

  Laser are also used to read and write data on optical storage media. including CDs, DVDs and Blu ray discs

- LiDAR

  LiDAR uses laser pulses to measure distance by calculating the time taken for light to travel to an object and return. It can be use to produce detailed three dimensional maps for surveying, enviromental studies and autonomous vehicle safety system (ADAS)



CO2 Laser Engraver



DJI Zenmuse L3 LiDAR Camera

Medical technologies

Applied physics plays a major role in modern medicine Many diagnostics and treatment methods depend on the interaction of radiation sound waves, magnetic Feild and Light with the human body. These technologies help doctors detect diseases and treat patient more accurately

Applications

Magnetic Resonance Imaging (MRI)

MRI uses a strong magnetic Feild to align the nuclear spins of hydrogen protons in a body. Radio Frequencies disturb these alignment, and the signal produced as protons return toward equilibrium are measured. Differences in relaxation times including T_1 and T_2, help produce detailed image of soft tissues

X Ray Computed Tomography

CT scanners take X-Ray measurements From different angles around the body. A computer process these measurements to reconstruct cross-sectional images. These images can also be combined to create three dimensional views of internal organs and structures



1.5 Tesla MRI machine



CT Scan Machine

Conclusion

Applied physics is an essential part of modern science and engineering because it connects theoretical knowledge with practical applications. Quantum mechanics has contributed to the development of advanced computing and semiconductor devices, while microelectronics has made electronic systems smaller and more powerful. Fiber optics enables fast communication over long distances, and laser technologies are used in industries, scientific research, and data storage. Similarly, medical technologies such as MRI, CT scans, ultrasound, and radiation therapy have improved the diagnosis and treatment of many diseases.

Overall, the study of applied physics helps engineers and scientists understand how modern technologies work and develop better solutions to practical problems. As technology continues to advance, the importance of applied physics will continue to grow in different areas of daily life.